

This is unreleased documentation for Serverpod **Next** version.

For up-to-date documentation, see the [latest version](#) (4.0.0).

3..16.12

RELATIONS

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Relations

A relation links rows in one table to rows in another through a foreign key, and the database keeps the link consistent. This is the alternative to embedding a model as a `json` column: related data lives in its own table, can be queried on its own, and is never duplicated per row.

Relations are declared in the model file with the `relation` keyword. There are two forms:

- An **object relation** types the field as the related model, e.g. `address: Address?, relation`. The related object can be fetched together with the row through [relation queries](#).
- An **id relation** stores only the foreign key, e.g. `addressId: int, relation(parent=address)`. You read and set the id yourself, and nothing else is fetched.

Choose a relation shape

Shape	Use when	Example on the page
One-to-one	A row pairs with at most one row of the other table.	A <code>User</code> with one <code>Address</code> .
One-to-many	One row owns any number of related rows.	A <code>Company</code> with many <code>Employee</code> rows.
Many-to-many	Rows on both sides connect freely, through a junction table.	<code>Student</code> and <code>Course</code> joined by <code>Enrollment</code> .
Self-relations	Rows relate to rows in the same table.	A <code>Cat</code> linking to its mother kittens.



Ask AI

Shape	Use when	Example on the page
Relations with modules	Your tables relate to a module's tables, such as the signed-in user.	A <code>UserProfile</code> attached to an <code>AuthUser</code> .

Whatever the shape, [Referential actions](#) control what happens to related rows when the row they point to is deleted or its id changes, and [Deferrable constraints](#) control when the foreign key is checked.

Related

- [Relation queries](#): fetch, attach, and detach related rows.
- [Filtering](#): filter on related rows, such as their count or contents.
- [Tables](#): how models map to tables in the first place.